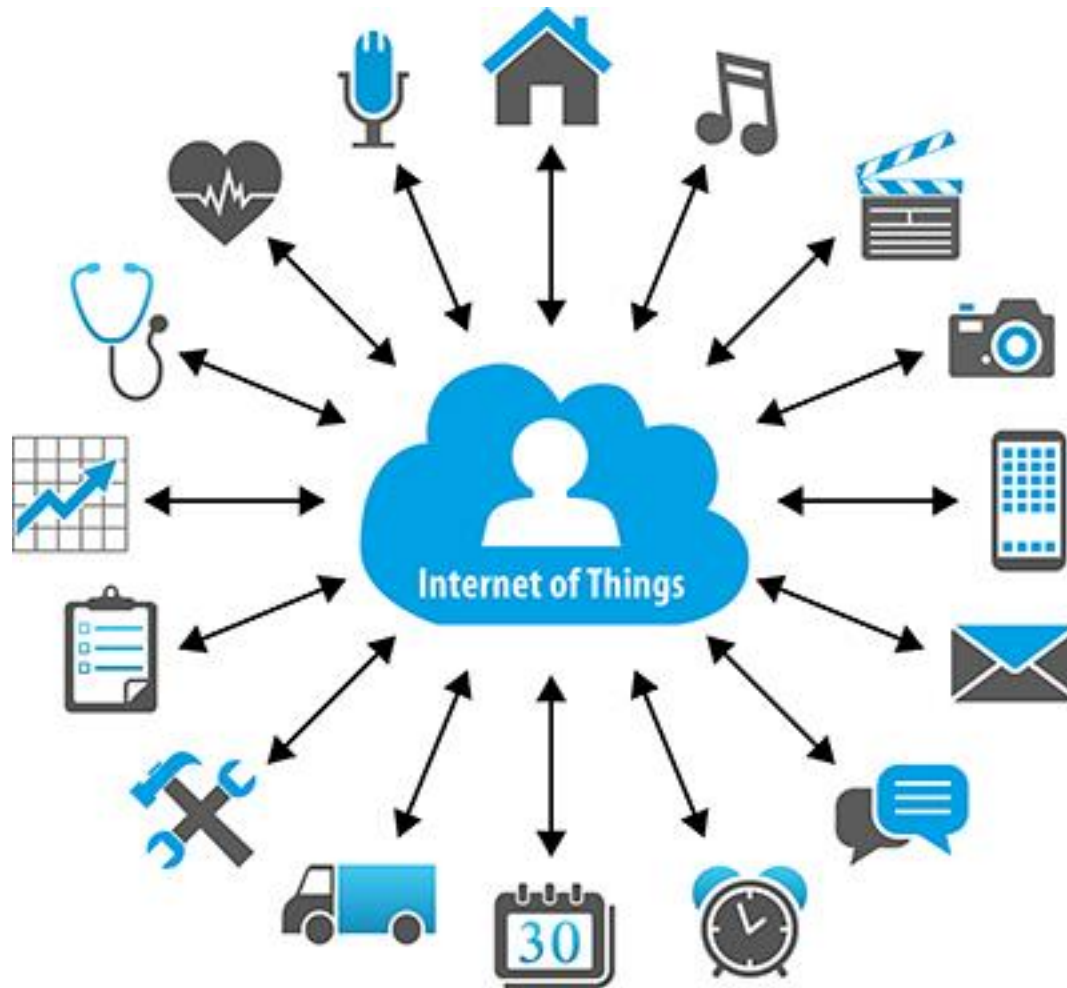




Introduction to IoT

Dr. Eman Elemam

Contents

- **IoT Definition.**
- **Evolution of IoT.**
- **IoT Architecture.**
- **Standardization Efforts.**
- **IoT Applications.**

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IoT Definition

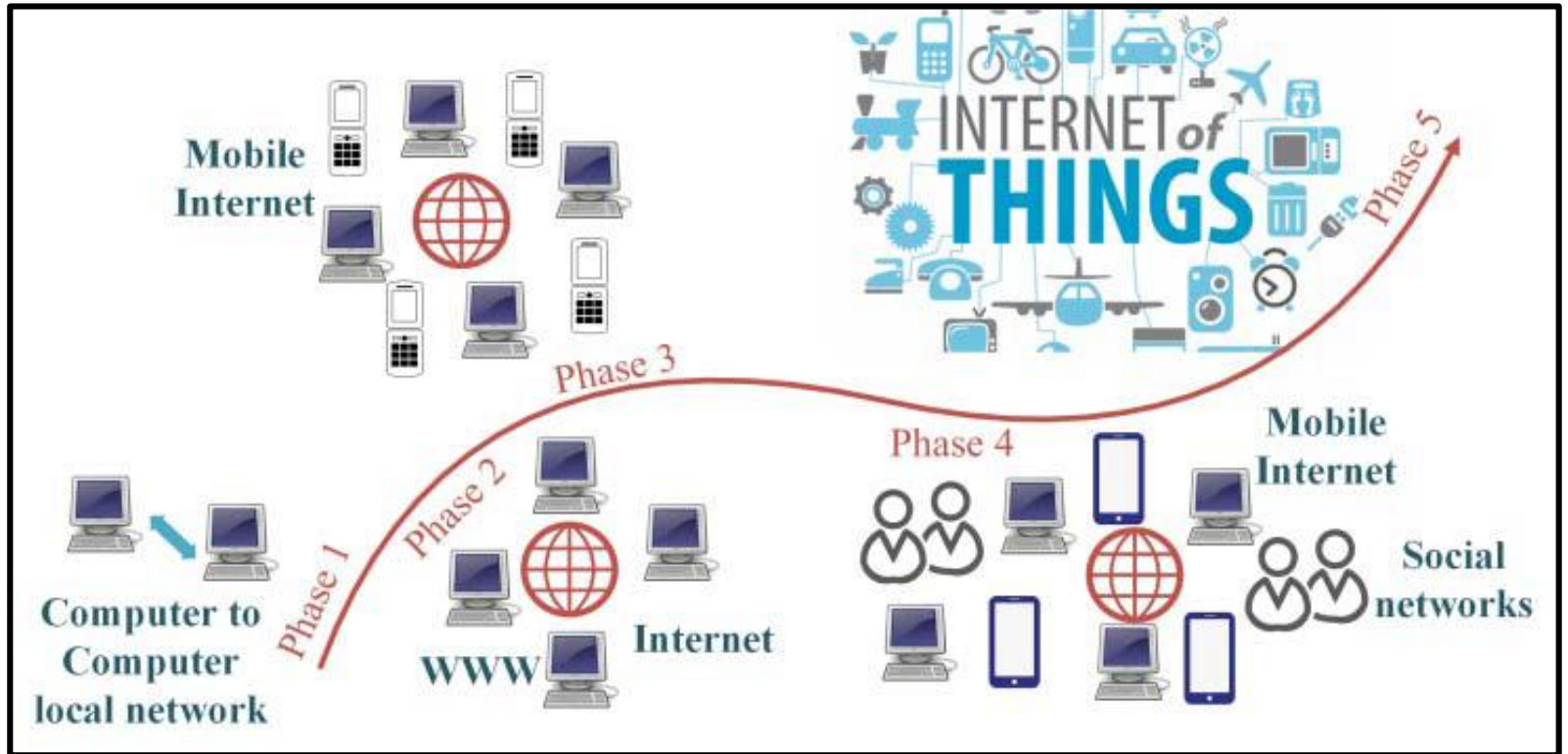
- IoT allows objects to be sensed or controlled remotely across existing network infrastructure, creating opportunities for more direct integration of the physical world into computer-based systems, and resulting in improved efficiency, accuracy and economic benefit in addition to reduced human intervention.
- 10 smart devices per capita is expected by 2025 as compared to two smart devices in 2015.



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Evolution of IoT



Evolution of IoT

- Computer networking began with the aim of economic and efficient sharing of expensive resources.
- The development of TCP/IP protocol suites lead to the advent of the Internet.
- The WorldWideWeb connected a large number of computers as a web.
- The mobile-Internet enabled mobile devices to be connected to the Internet.
- Peoples joined the Internet by the means of social networks.
- IoT allows people and things to be connected anytime, anyplace, with anything and anyone, ideally using any path or network.

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IoT Architecture

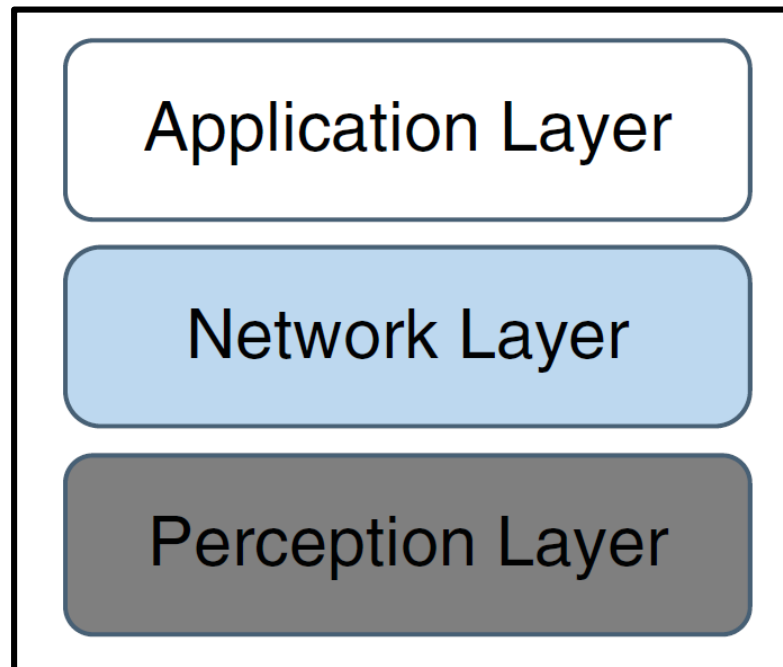
- No reference architecture model is yet firmly established.
- The architectural modeling of IoT is based on modifying the OSI standard with appropriate adjustments to the data link, network, and transport layers.



Three Layer Architecture

1. The Perception Layer (Device Layer):

- It interacts with physical objects using smart devices like RFID tags, sensors, actuators, etc.
- Its responsibilities include data acquisitions, primary information processing and transmitting the raw data or processed information to the upper layers.

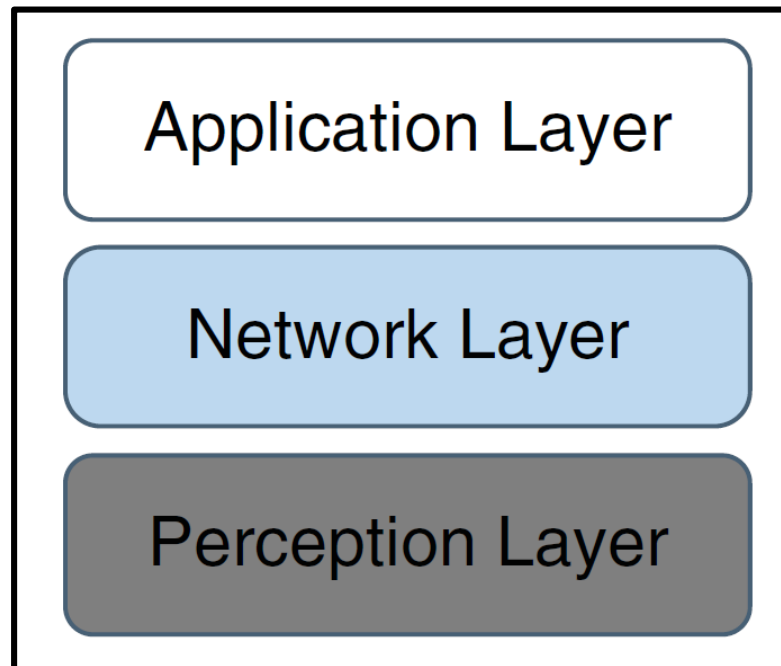


Three Layer Architecture

2. The Network Layer:

It enables optimal routing and data transmission through integration of heterogeneous networks using:

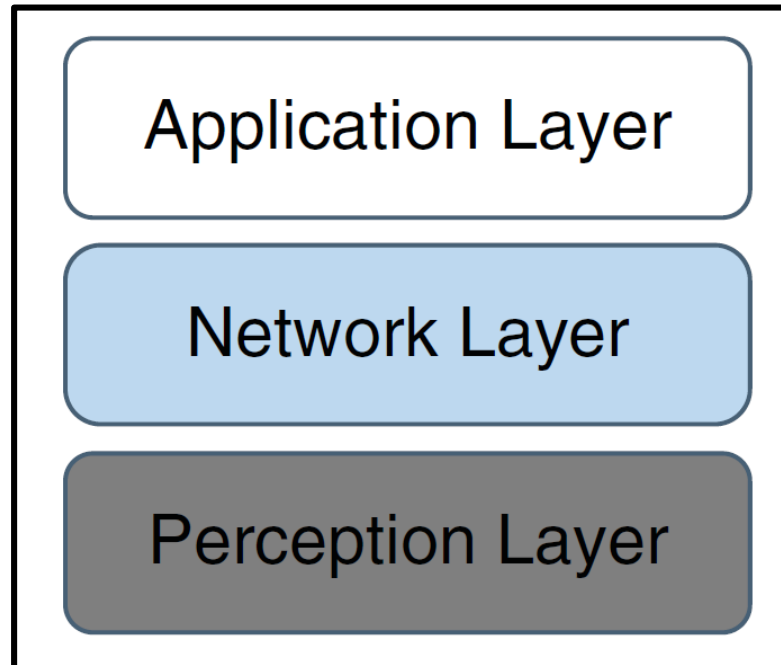
- Connecting devices (hub, switch, gateways, routers)
- Communication technologies (Bluetooth, WiFi, optical fibers, LTE)
- and Protocols (IEEE 802.15.4, 6LoWPAN, Zigbee, Z-Wave, CoAP, MQTT, XMPP, DDS).



Three Layer Architecture

3. The Application Layer:

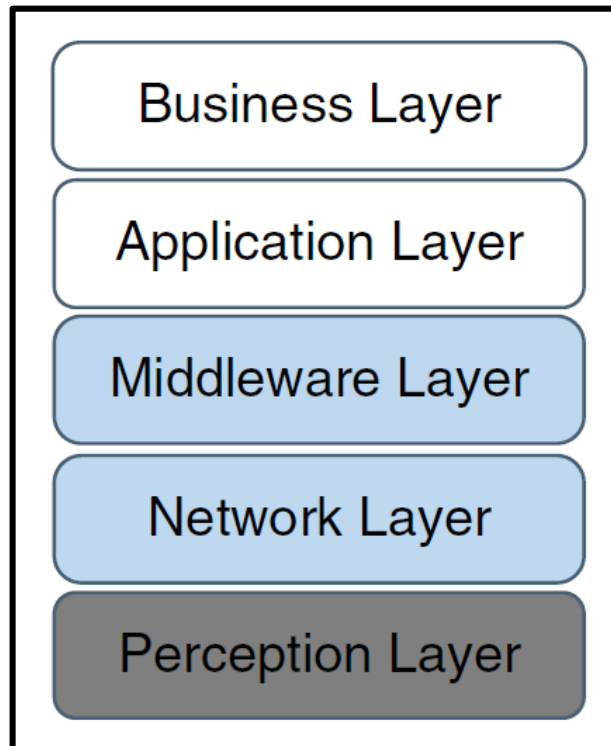
- It provides the essential services or operations to the users.
- This fulfills a combination of social and industrial demands in numerous domains including smart grid, smart transportation, smart cities, e-health, data services, etc.



Five Layer Architecture

4. The Middleware Layer:

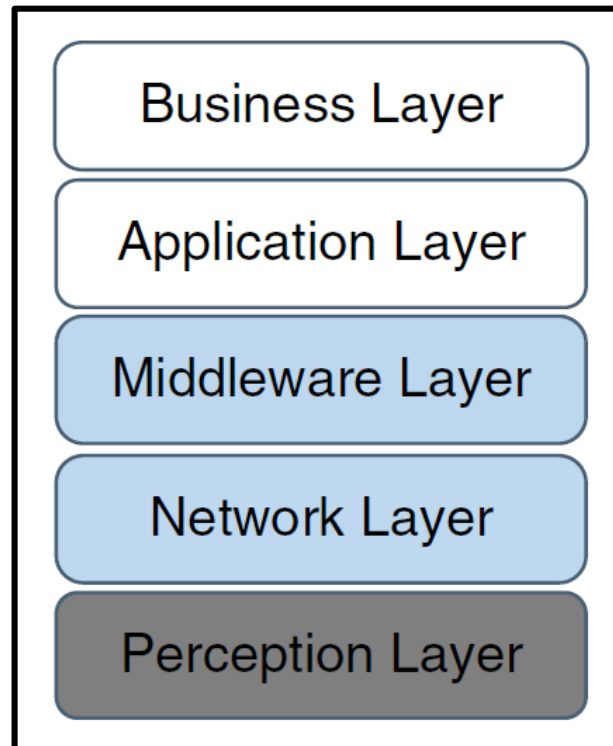
- It supports the service management between the network layer and the application layer.
- It receives information from the network layer, processes the information, performs computations and provides link to the database.



Five Layer Architecture

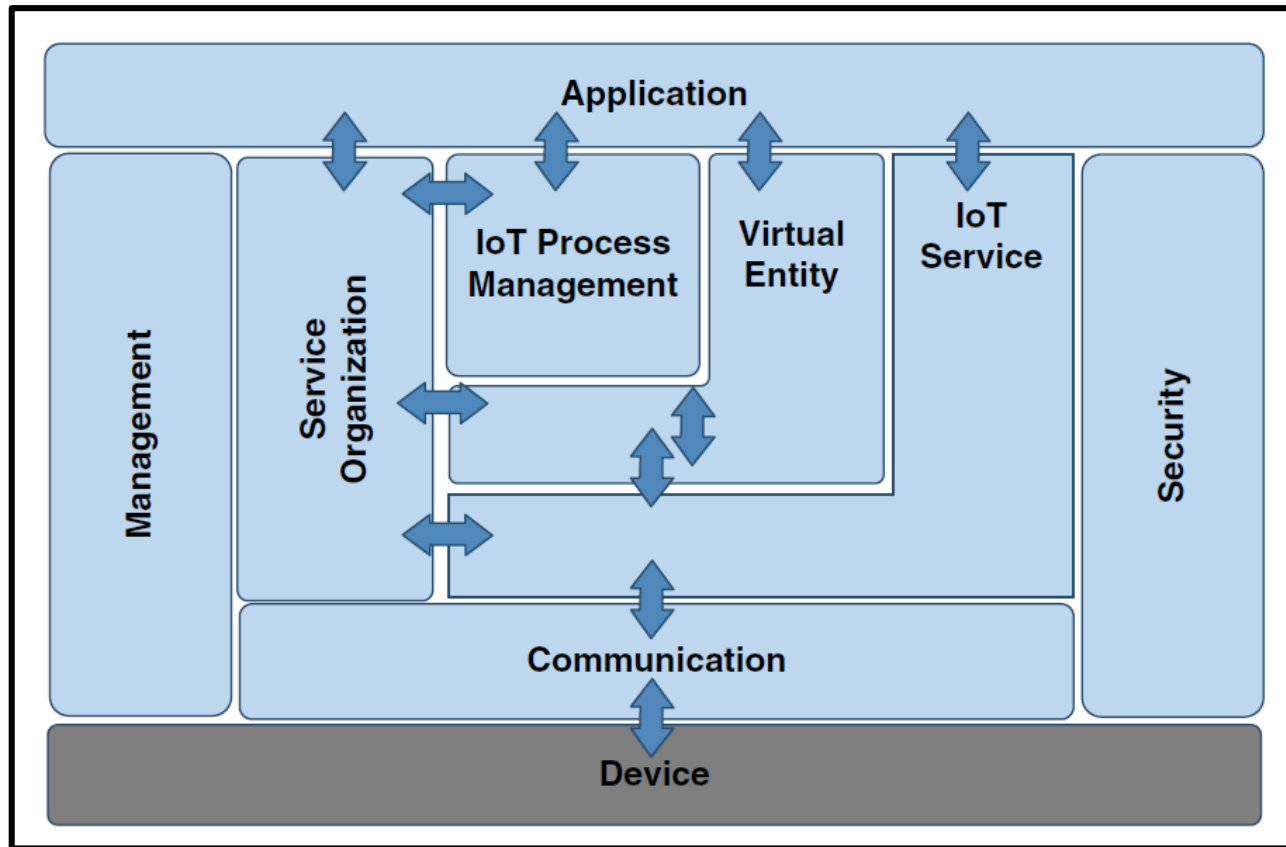
5. The Business Layer:

- It manages the overall IoT system.
- It determines the charging of various IoT applications.
- It builds the business models, graphs, flow charts, etc.



IoT Architecture (IoT-A)

- It concentrates on the generic aspects of informatics.



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Standardization Efforts

- Standardization of IoT has gained momentum since many organizations have stepped in and have boosted their contribution to develop a suite of protocols as well as open standards for IoT deployment that support inter-operable communication.
- Many organizations and communities have made contribution to consolidate IoT standardization activities like:
 - Internet Engineering task Force (IETF).
 - International Telecommunication Union (ITU).
 - Organization for the Advancement of Structured Information Standards (OASIS).
 - Institute of Electrical and Electronics Engineers (IEEE).

Application Layer Protocols

Applications protocols		RESTful	Transport	Publish/ Subscribe	Request/ Response	Security	QoS	Header Size (Byte)	Organization
	COAP	√	UDP	√	√	DTLS	√	4	IETF
	MQTT	X	TCP	√	X	SSL	√	2	OASIS
	MQTT-SN	X	UDP	√	X	SSL	√	2	OASIS
	XMPP	X	TCP	√	√	SSL	X	-	IETF
	AMQP	X	TCP	√	X	SSL	√	8	OASIS
	DDS	X	TCP UDP	√	X	SSL DTLS	√	-	OMG
	HTTP	√	TCP	X	√	SSL	X	-	IETF W3C

Network & Middleware Layers Protocols

Network layer & middleware protocols		Functionality	Characteristics	Organization
	mDNS	Service discovery	Zero configuration. Uses IP multicast UDP packets. Run with no or less infrastructure	IETF
	DNS-SD	Service discovery	Utilizes mDNS. Zero configuration. Wide area service discovery by clients.	IETF
	RPL	Routing	Distance vector protocol. DODAG topology.	IETF
	6LoWPAN	Encapsulation	Adaptation layer for IPv6 . Header compression. Fragmentation.	IETF
	6TiSCH	Encapsulation	Low-power operations for IPv6 over IEEE802.15.4e TSCH mode. Industrial IoT.	IETF

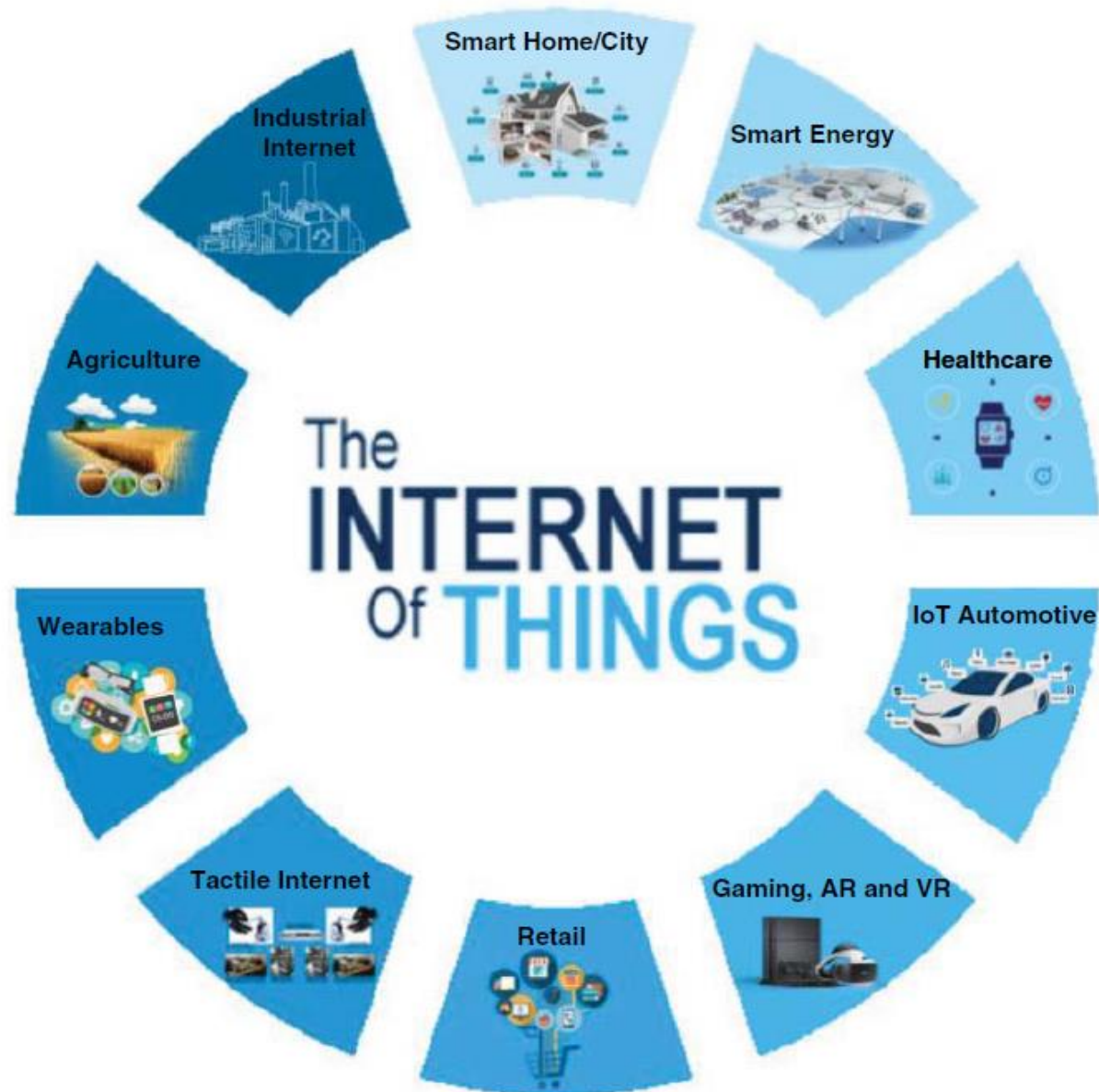
Physical Layer Protocols

Physical layer protocols		Spreading Technique	Radio Band (MHz)	MAC Access	Data Rate (bps)	Scalability	Organization	Range
	IEEE 802.15.4	DSSS	868/915/2400	TDMA, CSMA/CA	20/40/250K	65K nodes	IEEE	10 m-100m
	BLE	FHSS	2400	TDMA	1024K	5971slaves	Bluetooth group	< 100m
	EPCglobal	DS-CDMA	860~960	ALOHA	Varies5~640K	-	EPCglobal	< 50 m
	LTE-A	Multiple CC	varies	OFDMA	1G (up), 500M (down)	-	3GPP	< 100km
	Z-Wave	-	868/908/2400	CSMA/CA	40K	232nodes	Sigma Designs	30 m (Indoor) & 100m (Outdoor)
	LoRa	LoRa (CSS)	0.125-0.25	Unslotted ALOHA	0.3K to 50K	Upto millions	LoRa Alliance	< 15 km
	Sigfox	BPSK	0.1	Unslotted ALOHA	100or 600	Upto millions	SNO	30 km – 50 km (Rural) & 10 km (Urban)
	NB-IoT	QPSK	0.18	FDMA/OFDMA	20K(up) 250K(down)	50K	3GPP	< 35 km
	Insteon	-	904	TDMA + simulcas	38.4k	256	Smartlabs	< 45 m

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IoT Applications



Categorization of IoT Applications

IoT Application	Classification According to Number of Devices	Classification According to Scope of Usage
Smart Home	Massive IoT	Individual level
Smart City	Massive IoT	Infrastructural level
Smart Energy	Massive IoT	Infrastructural level
IoT Automotive	Critical IoT	All-inclusive level
Healthcare	Massive IoT	All-inclusive level
Gaming, AR, VR	Massive IoT	Individual level
Retail	Massive IoT	Organizational level
Wearable	Massive IoT	Individual level
Smart Agriculture	Massive IoT	Organizational level
Industrial Internet	Critical IoT	Organizational level

Classification According to No. of Devices

1. Massive IoT:

- Large numbers of low-cost low-powered devices typically emit a low volume of non-delay sensitive data.
- The devices need to report to the cloud on a regular basis
- Therefore require seamless connectivity and good coverage.

2. Critical IoT:

- Have very high demands of reliability, availability, and low latency.

Classification According to Scope of Usage

1. Infrastructural Level:

- They have potential, in turn, to create the next level of the ecosystem.

2. Organizational Level:

- Aim to automate the working of an organization.

3. Individual Level:

- Personal usage.

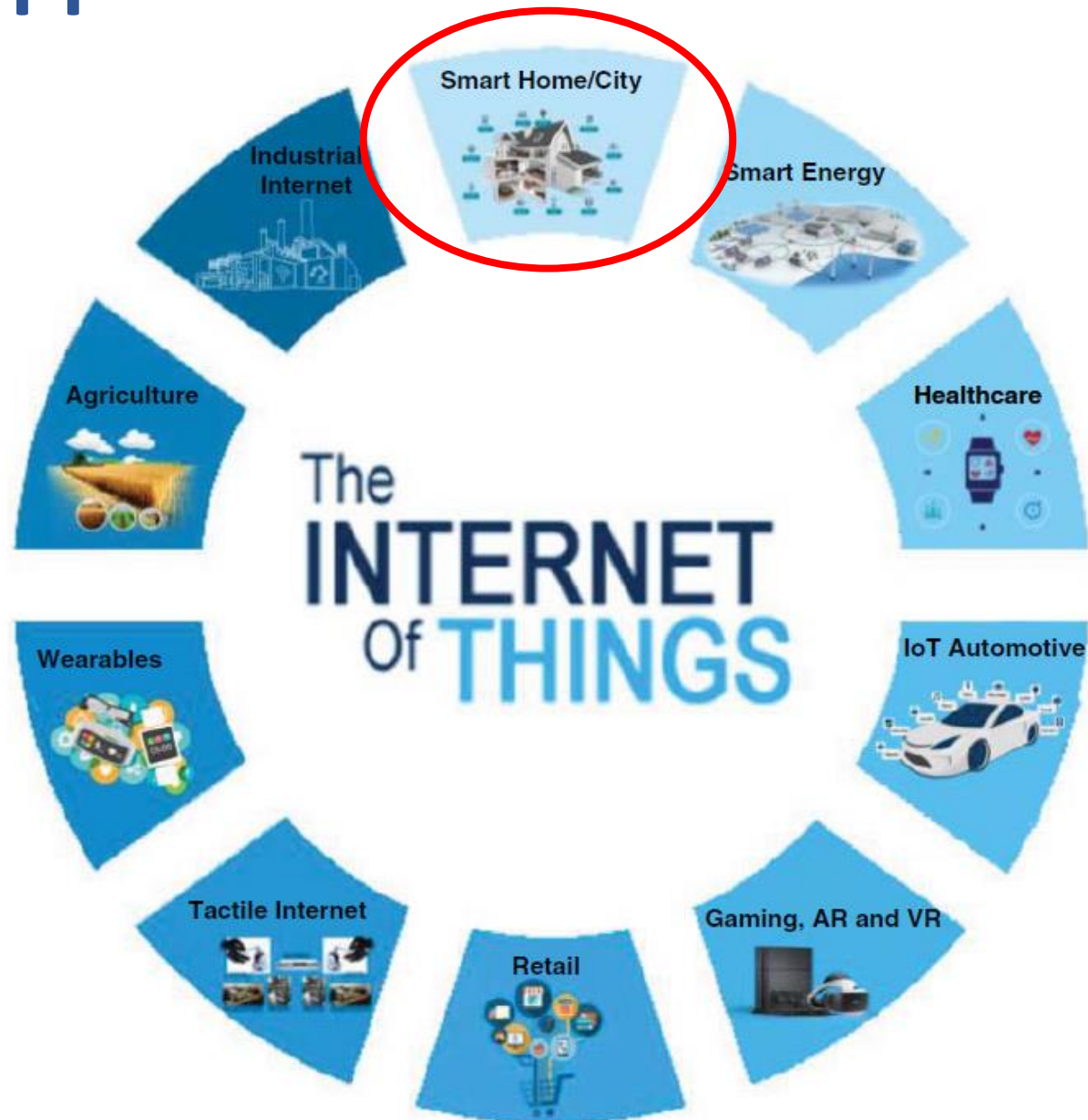
4. All-inclusive Level:

- Have wider scope and can span through all the levels.

Characteristics of IoT Applications

IoT Application	Data type	Backhaul Connectivity	Expected latency
Smart Home	Stream / Historical data	Realtime	1 ms -1000 s
Smart City	Stream / Massive data	Realtime	$\leq 1\text{ms}$
Smart Energy	Stream / Massive data	Realtime/ Intermittent	1ms - 10 mins
IoT Automotive	Stream / Massive data	Realtime	$\leq 1\text{ms}$
Remote surgery	Stream data	Realtime	$\leq 200\text{ ms}$
Remote consultancy	Stream data	Realtime	1 ms-100 s
Gaming, AR, VR	Stream / Massive data	Realtime	$\leq 1\text{ms}$
Retail	Stream / Historical data	Realtime/ Intermittent	$\leq 1\text{ ms}$
Wearable	Stream data	Intermittent	Several Hours
Smart Agriculture	Historical data	Intermittent	Several hours
Industrial Internet	Stream / Massive data	Realtime	$\leq 1\text{ ms}$
Tactile Internet	Stream	Realtime	$\leq 1\text{ ms}$

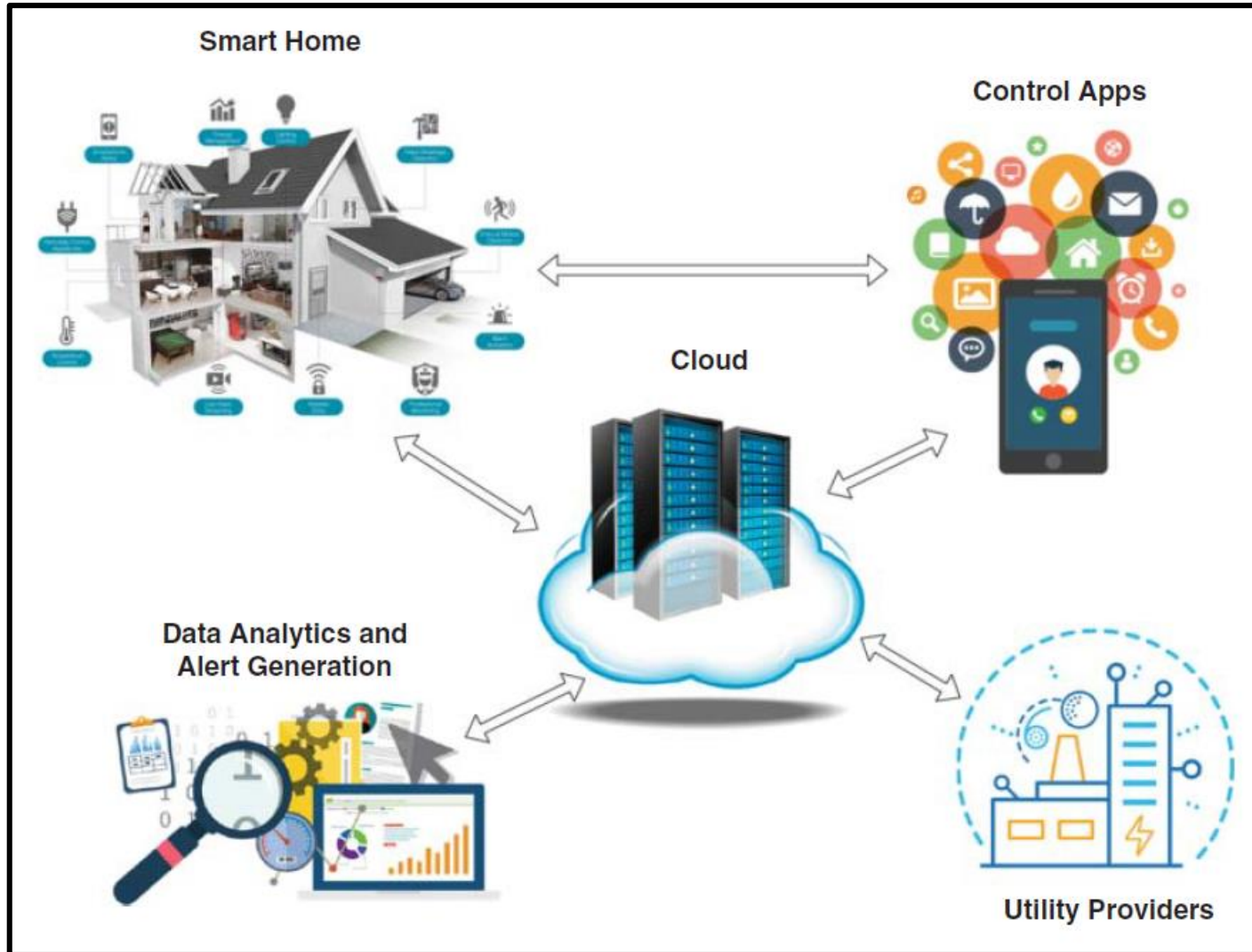
IoT Applications



1. Smart Home

- It is used to monitor, manage, optimize, and access remotely the home environment comprising home appliances while minimizing human effort.
- It aims to optimize the energy consumption taking into account the usage pattern and real-time presence of residents, the external environment (e.g. weather condition), time of the day, balance units of pre-paid electricity account, etc.
- It makes use of both local (limited) storage (EX: gateway or hub) as well as cloud infrastructure.
- The achieved gain would be in terms of latency, load balancing, traffic reduction and progressive-resource utilization.

Smart Home Stakeholders



Smart Home Components

- Smart security & surveillance systems.
- Smart HVAC (Heating, Ventilation and Air Conditioning).
- Self-adjustable and smart customization of the environment based on the user's profile.
- Smart energy management.



Smart Home Technologies & Challenges

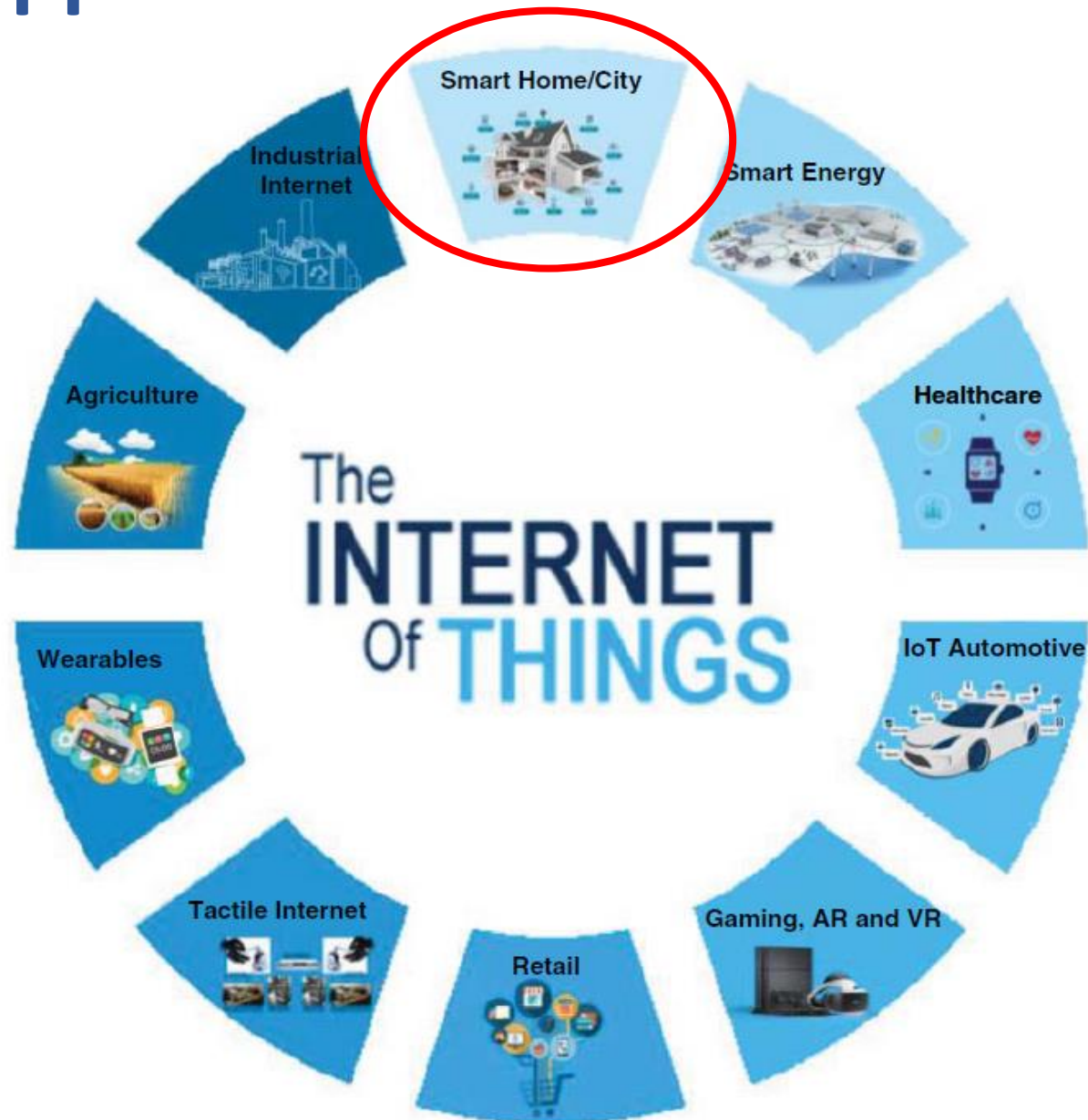
➤ Technologies & Standards:

- ZigBee.
- 6LoWPAN.
- RFID
- Cloud computing.

➤ Challenges:

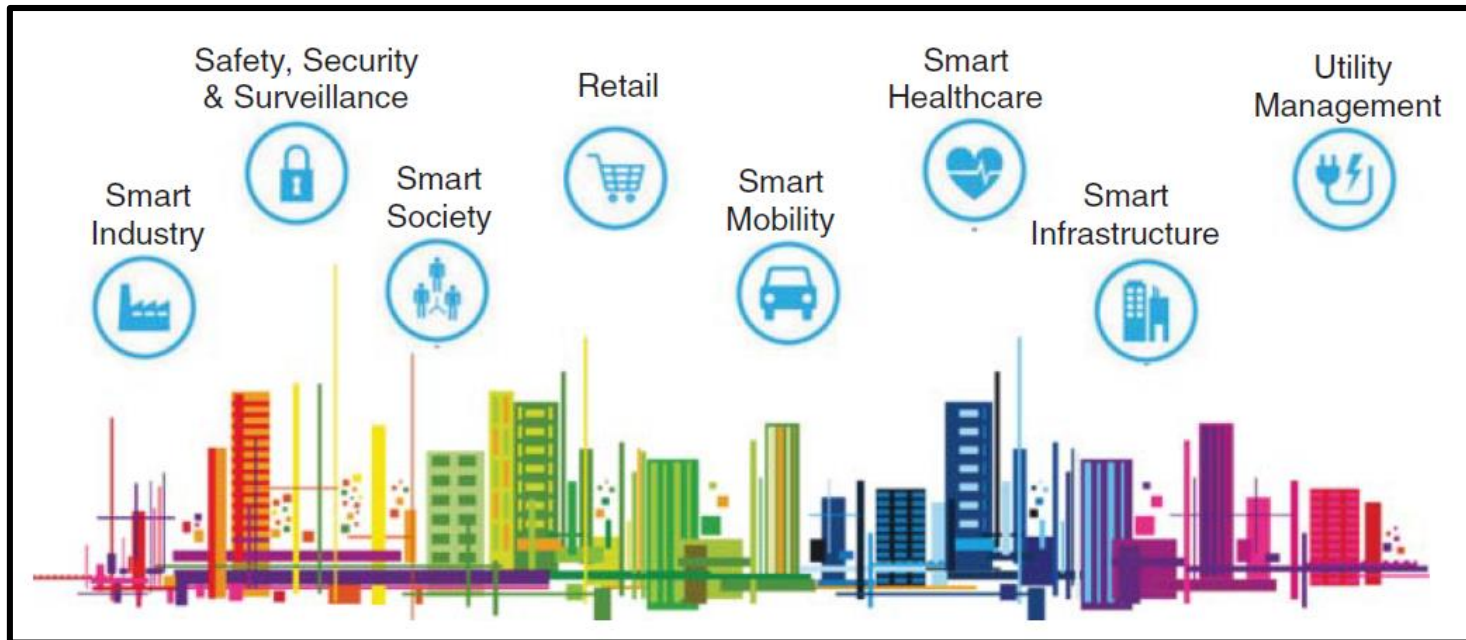
- High privacy and security.
- High reliability.
- High interoperability.
- Strong adaptation to multipath error prone wireless environment.

IoT Applications



2. Smart City Components

- Smart hygiene.
- Smart mobility & traffic management.
- Smart governance.
- Smart development of infrastructure.
- Smart utility management.



Smart City Technologies & Challenges

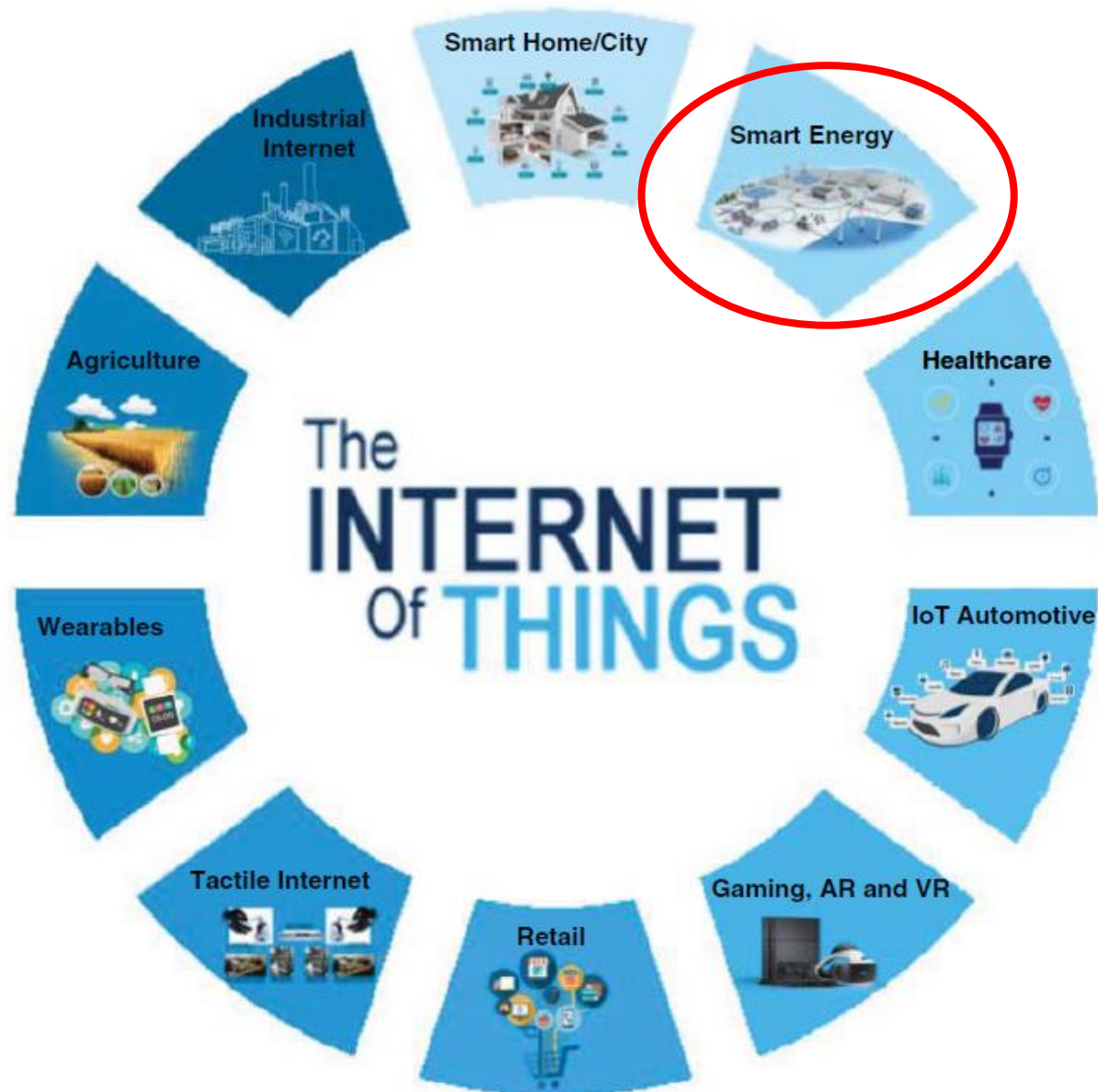
➤ Technologies & Standards:

- Small range connectivity: ZigBee, RFID, Bluetooth, Wi-Fi.
- Long Range Connectivity: GSM, WCDMA, 3G, LTE and 5G.

➤ Challenges:

- High scalability.
- Low latency.
- High reliability.
- High availability.
- High security.

IoT Applications

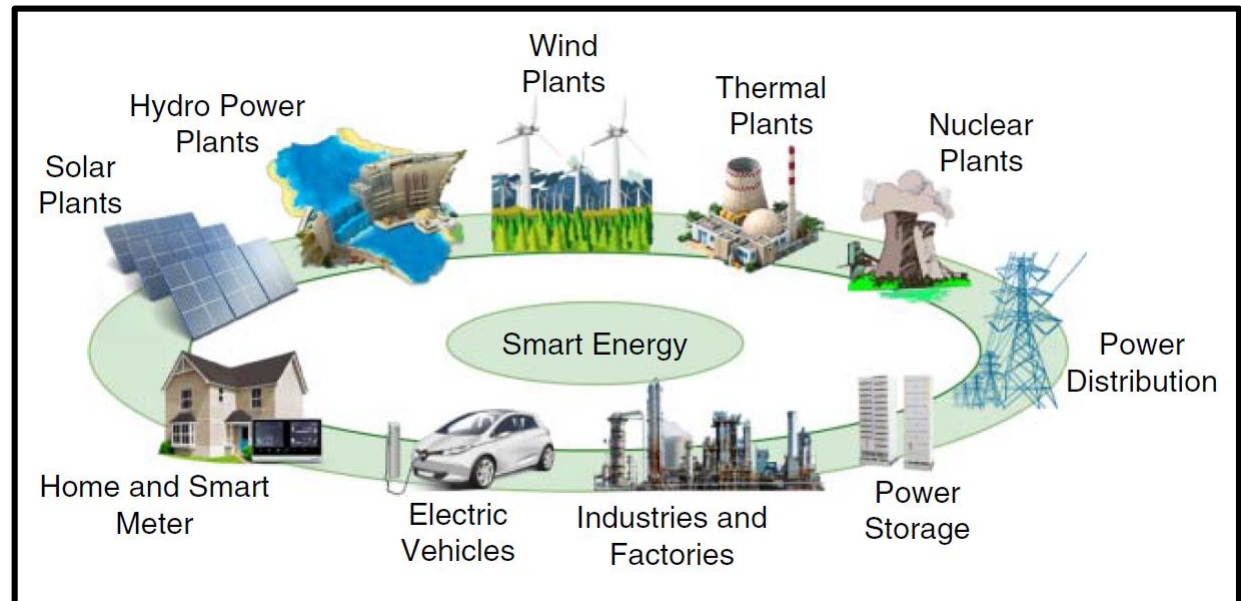


3. Smart Energy Tasks

- Power generation at the source.
- Transmission of power over high-voltage lines from source up to substations.
- Distribution to the end consumers.
- Billing at predefined time cycle.
- 24x7 monitoring & maintenance comprising of fault detection and rectification.

Smart Energy System's Components

- Smart energy generation including renewable energies.
- Smart maintenance involving diagnosis, early prediction of failure, and proactive rectification.
- Smart access to real-time usage to optimize consumption and for billing purposes.
- Smart capacity building based on the conclusions drawn from data analytics applied over huge data collected from the users.



Smart Energy Technologies & Challenges

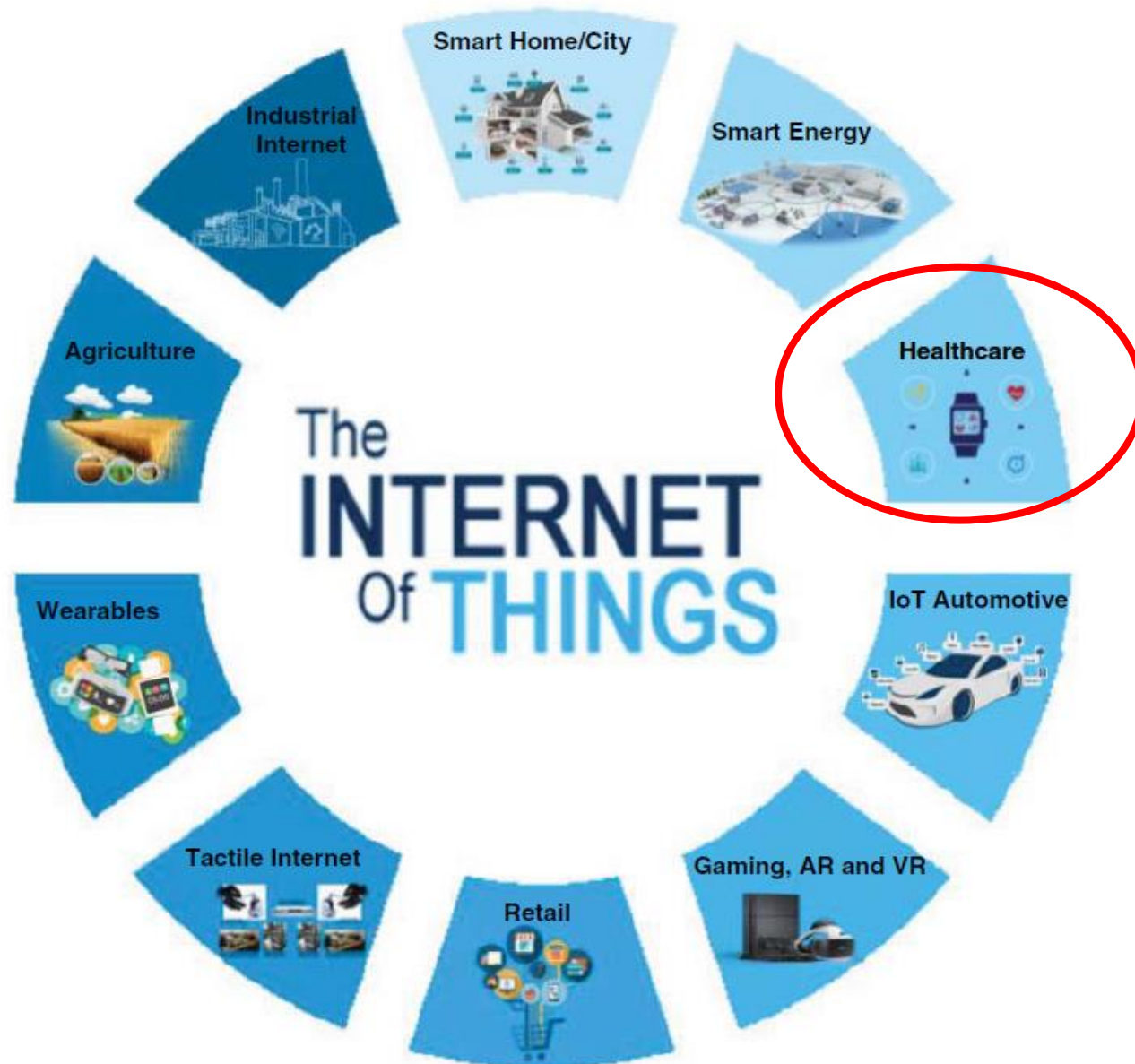
➤ Technologies & Standards:

- LoRa.
- Power-Line Communication (PLC).
- Z-Wave.

➤ Challenges:

- High sustainability.
- High resiliency.
- High safety.
- High level of optimization.

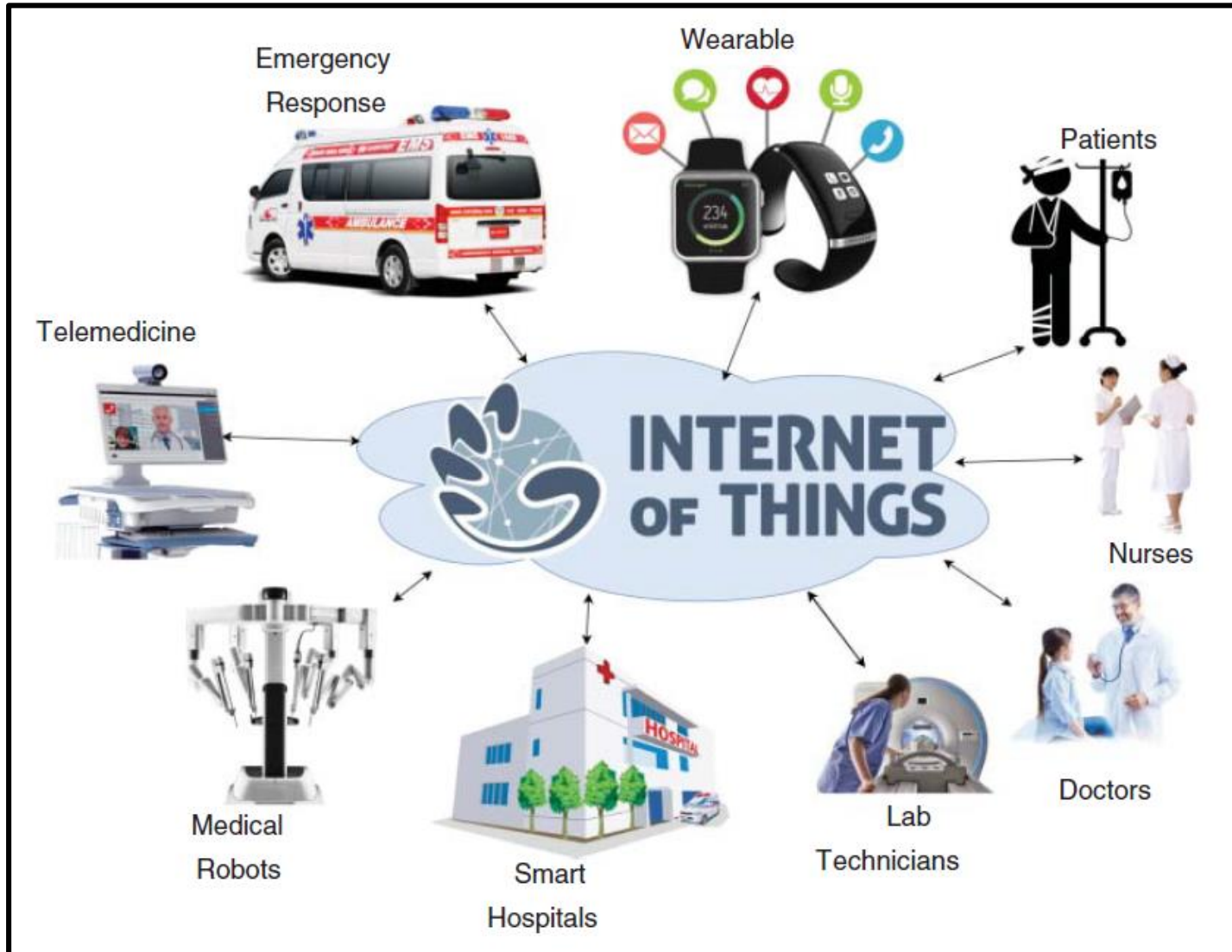
IoT Applications



4. Healthcare

- It is envisioned to capture the biggest chunk of the future IoT market by 2025.
- The IoT paradigm can monitor various physiological parameters, detect symptoms and thereby (early) diagnose, suggest preventive measures, and progressively adapt treatment based on AI and ML approaches.
- Medical IoT can guide pharmaceutical companies to develop new medicines based on data analytics of recent patients.

Healthcare Stakeholders



Healthcare Components

- Smart remote health monitoring.
- Smart asset management for hospitals.
- Smart medical inventory optimization based on real-time healthcare data analytics.
- Smart patient-doctor rapport.
- Smart surgeries.



Healthcare Technologies & Challenges

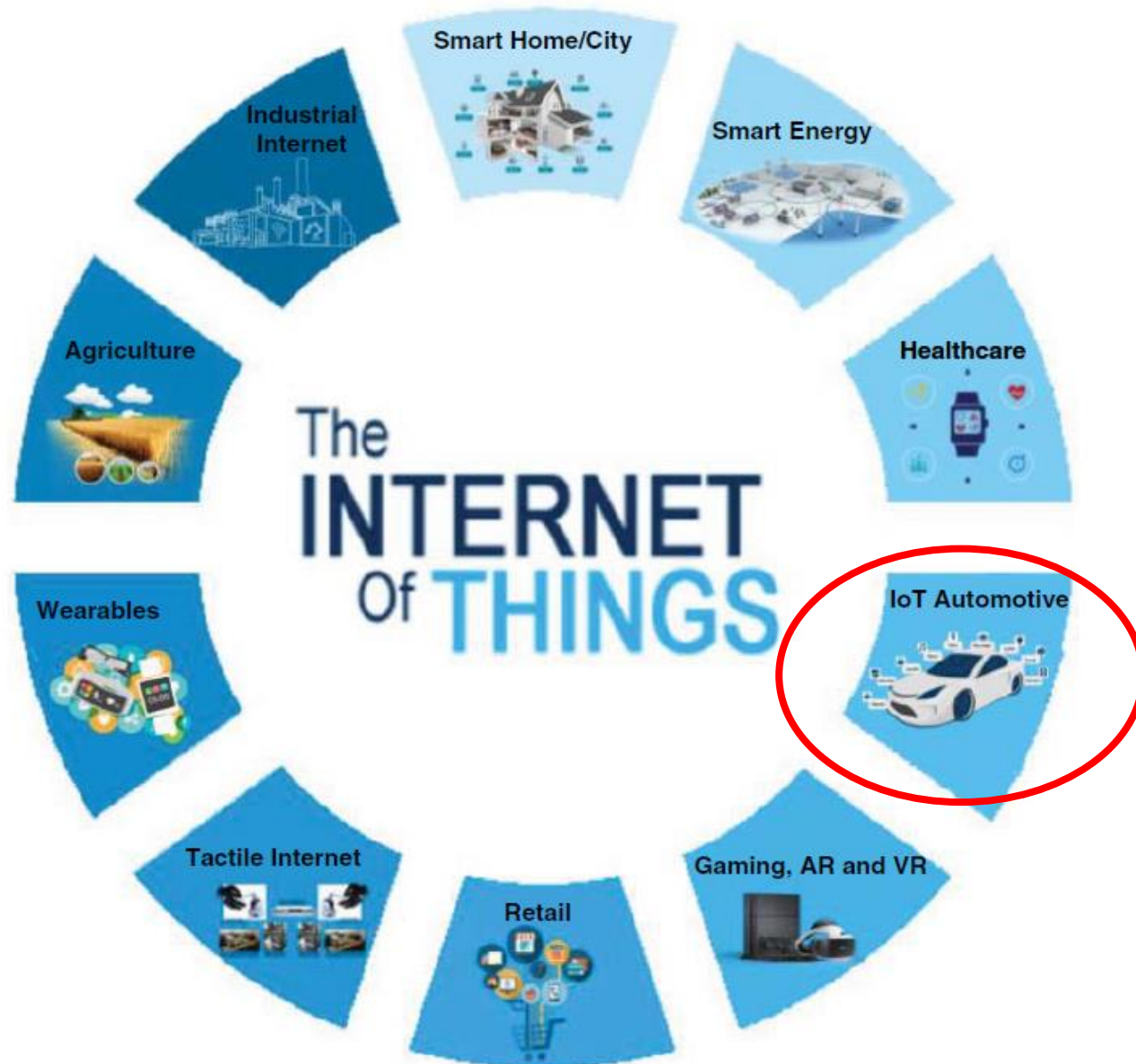
➤ Technologies & Standards:

- BLE.
- WBAN (IEEE 802.15.6).
- LR-WPAN (IEEE 802.15.4).
- NB-IoT.

➤ Challenges:

- Ultra-high safety.
- Ultra-high precision.
- High trustworthiness.
- High privacy
- Low energy consumption.

IoT Applications



5. IoT Automotive

- Many companies, like Tesla, BMW, Ford, are working in the direction of highly automated (level - 4) or fully automated (level - 5) vehicles.
- Today, there are approximately 60 to 100 sensors embedded in a single vehicle and soon the number will rise to 200.
- Drones or UAV (Unmanned Aerial Vehicles) acting as sensor devices open-up avenues for IoT applications in agriculture, mining operations, and industrial inspection services.



Automotive Components

- Smart self-optimization and maintenance to be carried out by the vehicle itself or remotely by the owner.
- Smart security and safety systems.
- Smart customization of ambience based user's profile.
- Smart navigation.



Automotive Technologies & Challenges

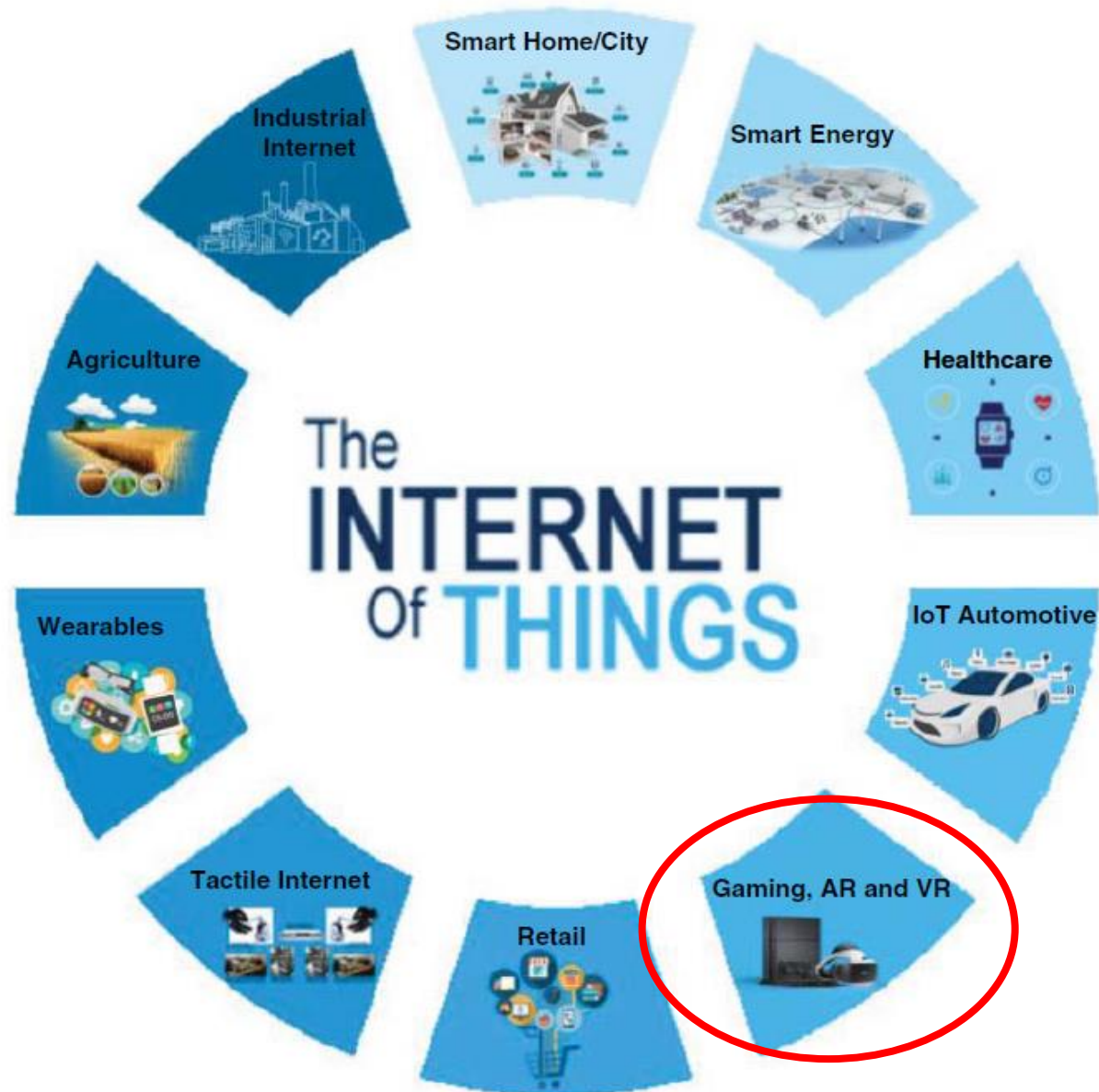
➤ Technologies & Standards:

- Bluetooth.
- ZigBee.
- Dedicated Short Range Communication (DSRC).
- 4G cellular technology.

➤ Challenges:

- Low power.
- High privacy.
- High bandwidth.
- Required minimization of roaming issues.
Consequently, minimizing the latency and handling the fluctuations in the price of home vs. foreign network providers.

IoT Applications



6. Gaming, AR, VR and MR

- Augmented Reality (AR) is usually accessed usually with a smartphone's camera and it involves overlaying visual elements, sound, and other sensory stimuli onto a real-world to enhance the user experience.



Gaming, AR, VR and MR

- Virtual Reality (VR) is a computer-generated environment with scenes and objects that appear to be real, making the user feel they are immersed in their surroundings.



VR vs. AR



AUGMENTED REALITY IS
COMPUTER GENERATED
GRAPHICS ON TOP OF
ACTUAL REALITY

VS.



VIRTUAL REALITY IS
COMPUTER GENERATED
GRAPHICS TO CREATE AN
ENTIRE VIRTUAL WORLD

Gaming, AR, VR and MR

- Mixed Reality (MR) experience combines elements of both AR and VR, where real-world and digital objects interact.



Gaming, AR, VR and MR

- The purpose behind gaming is no more confined merely to entertainment. Thus, gamification is yet another fascinating area which has been explored in various sectors like education, training, & health management.



Gaming, AR, VR and MR Applications



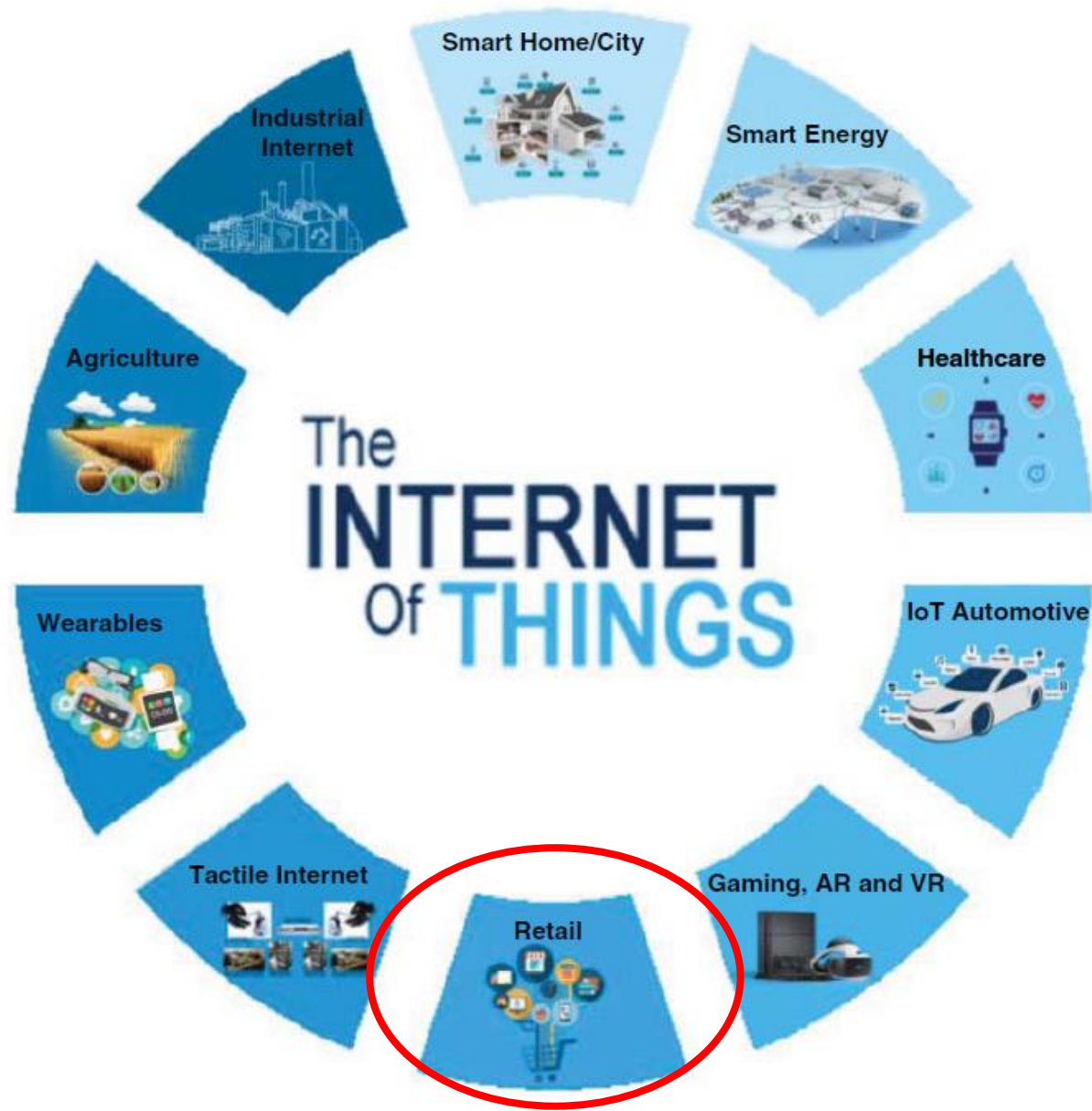
VR Astronaut Training



Metaverse Challenges

- Real-time connectivity.
- High bandwidth.
- Low power consumption.
- Ultra-low latency and interactive response time that essentially bridges the gap between virtuality and reality in order to minimize cyber-sickness.

IoT Applications



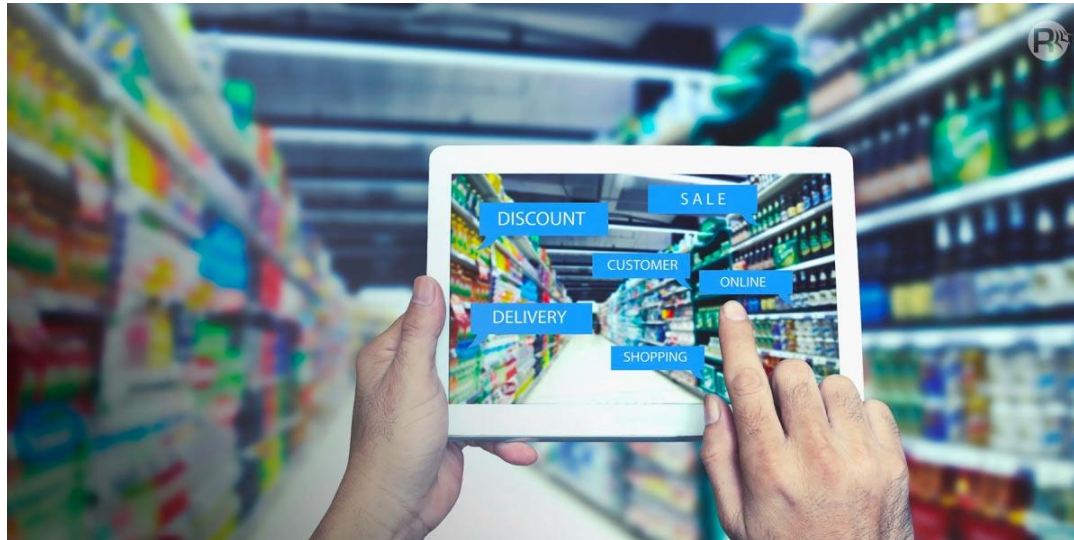
7. Retail

- Entire new sets of real-time services can be introduced that at one end enhances the customer experience and alleviates the way the retail sector is managed and maintained.

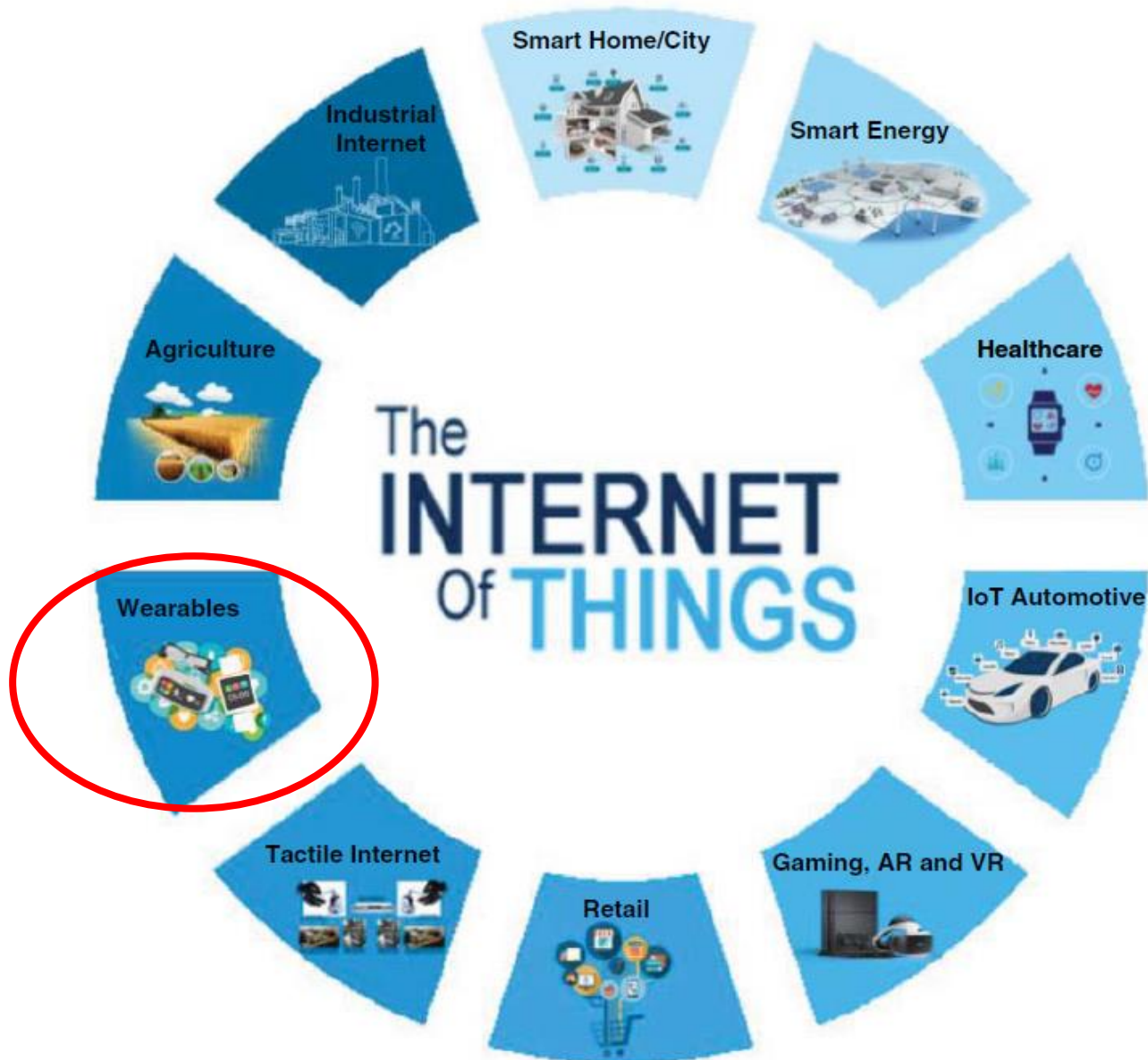
- **Challenges:**
 - Aesthetic presence.
 - Customer friendliness.
 - Context awareness.
 - High QoE of customers.

Retail Components

- Smart supply chain and logistics.
- Smart finance management
- Smart real-time customer assistance while purchasing.
- Smart complains management system.
- Smart post-purchase relations and feedback system.



IoT Applications



8. Wearable

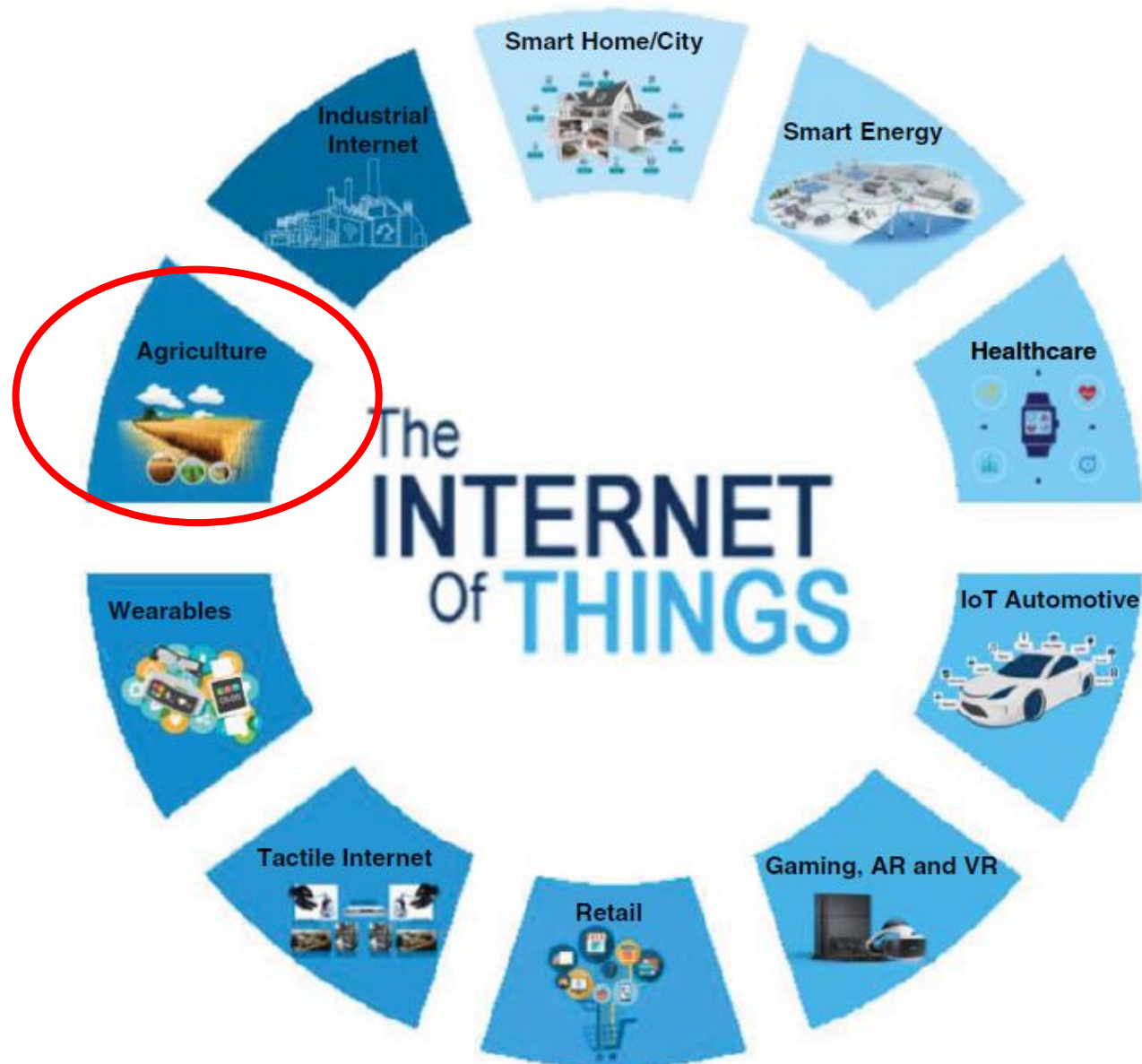
- Wearable IoT devices are closely interwoven in our present life style. Ranging from trendy smart watches, to essential medical devices.
- They hit the market in a big way and they will stand just next to smartphones in consumer electronics.
- **Challenges:**
 - Ultra-high safety.
 - Ultra-low power consumption.
 - High level of comfort to the human body.
 - Highly user-friendly.
 - Low latency.
 - High context awareness.
 - High privacy.
 - High bandwidth.

Wearable IoT Components

- Smart tracking.
- Smart clothing.
- Smart medical monitoring.
- Personal security alarms.



IoT Applications



9. Agriculture Tasks

- Monitoring and acquisition of ambient data which is of a large variety and of huge volume.
- Aggregation of data and exchange over the network.
- Data analysis and decision making.
- It may help in the prediction of yield to ensure the economic value to be gained.
- Early detection of diseases spread in crops to enable timely preventive measures.

Agriculture Components

- Smart plowing.
- Smart seedbed preparation and planting.
- Smart irrigation.
- Smart fertilization.
- Smart harvesting,
- Smart livestock management which deals with smart animal tracking, smart health monitoring, smart feed and fodder management.



Agriculture

➤ Technologies & Standards:

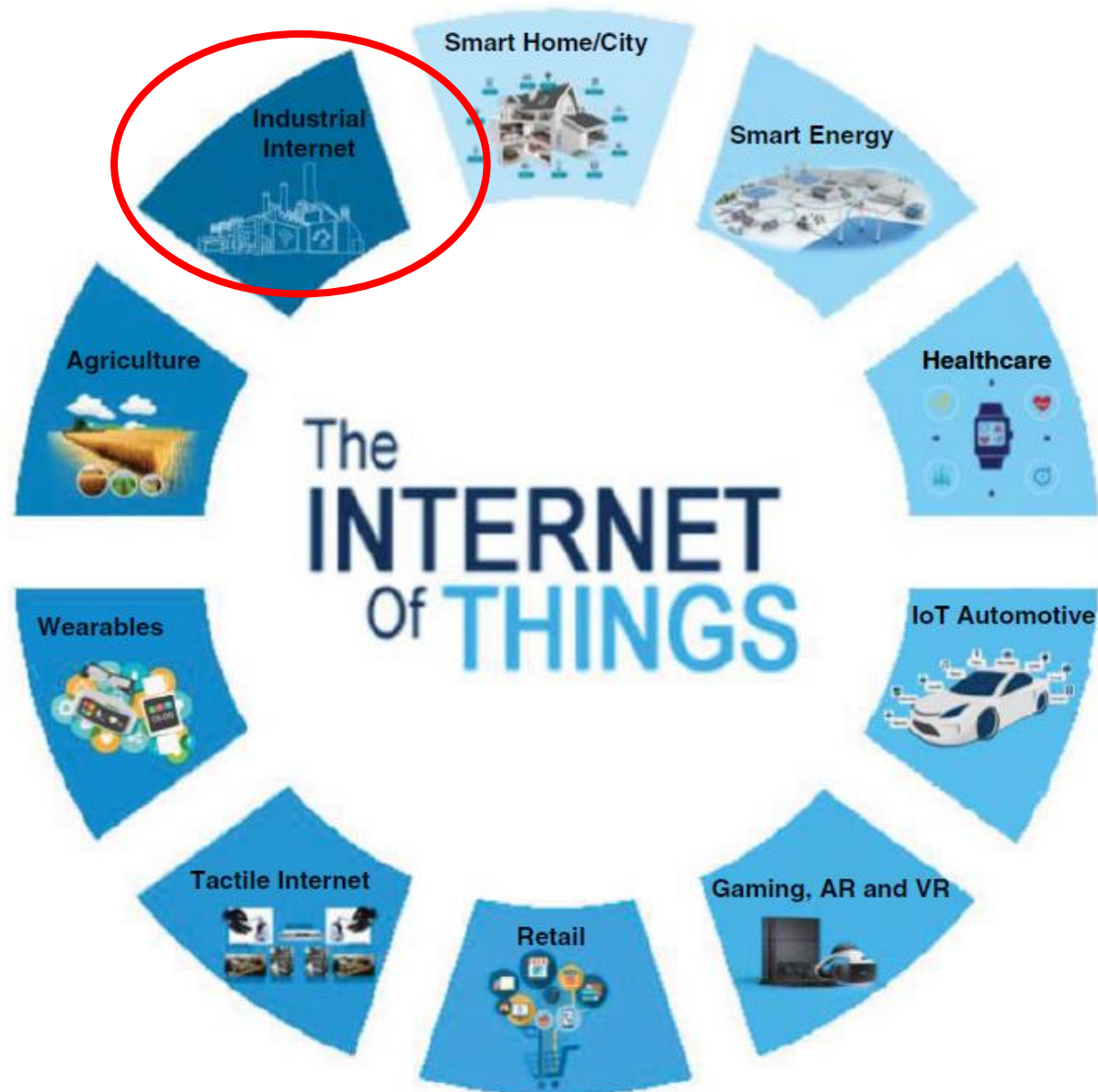
- Bluetooth.
- ZigBee.
- RFID.
- GPS.
- LoRa.
- NB-IoT.
- Cloud computing.

➤ Challenges:

- Low-cost.
- Low-power.
- Highly reusable.
- Highly efficient (big) data management.
- Scalable and progressively extendable solution.



IoT Applications



10. Industrial Internet

- An industrial application of IoT is referred as IIoT.
- IIoT securely interconnects industrial assets over the Internet with the use of wide varieties of technologies.
- IIoT leverage relevant technologies like cloud computing, edge and fog computing, big data analysis, AI and ML.
- IIoT has huge potential to facilitate precise supervision in an industrial environment, optimize value of productivity at low cost of operations and maintenance and thus, leads to an increase in the return on investment for stakeholders.

Industrial Internet

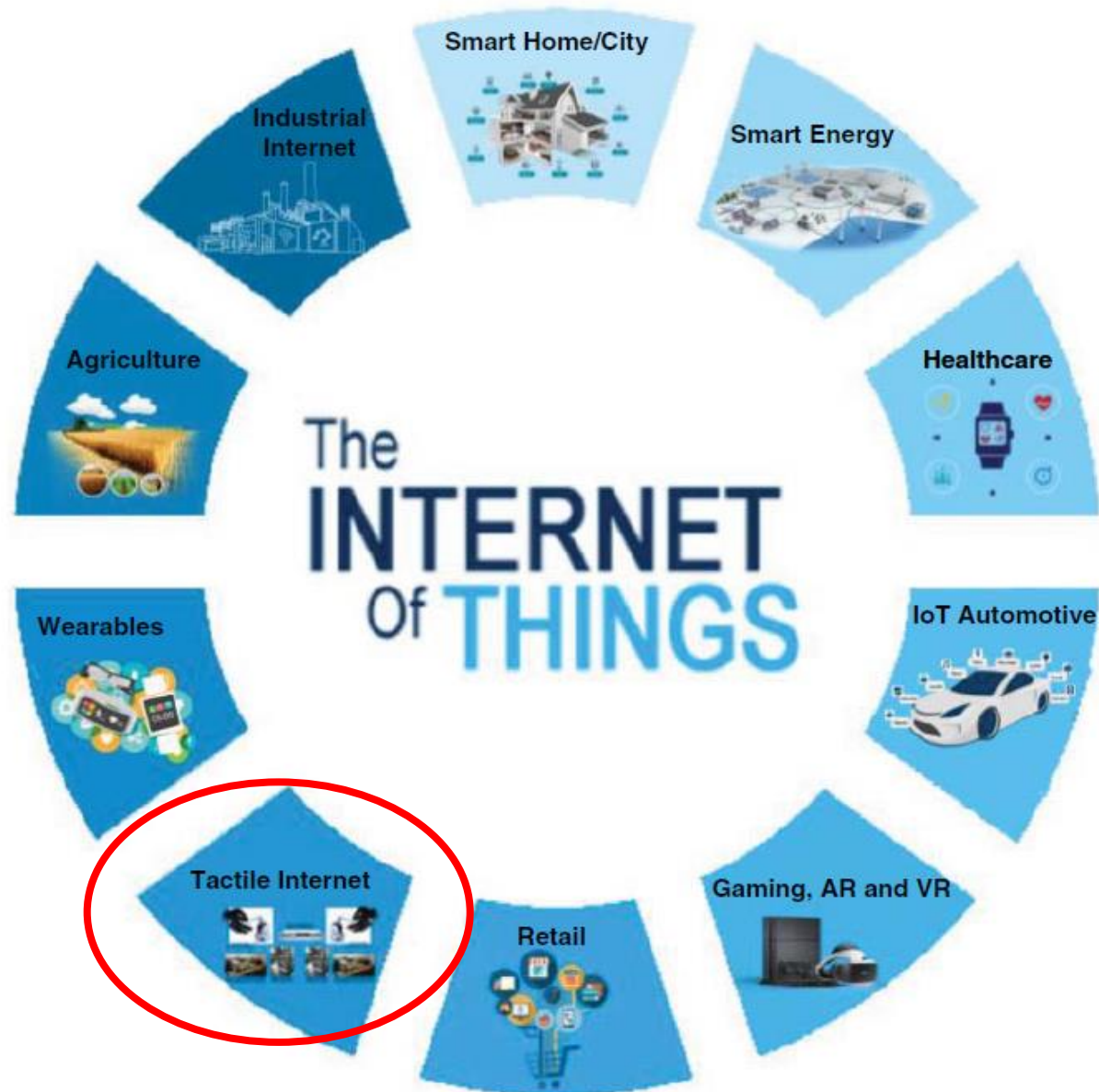
➤ **Components:**

- Smart manufacturing.
- Smart customer relationships.
- Smart supply chain.
- Smart budgeting.
- Resource management.

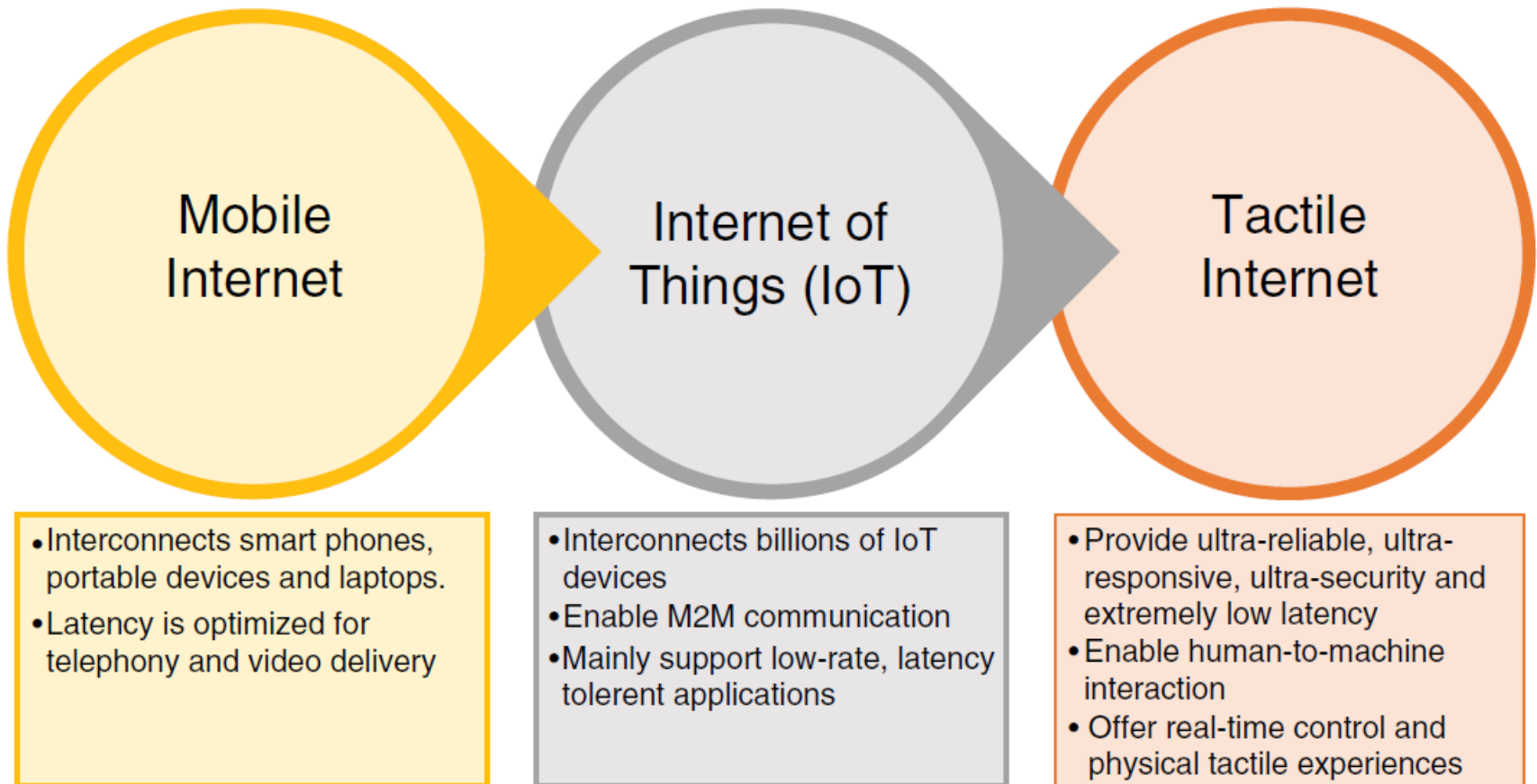
➤ **Challenges:**

- Low latency.
- High precision.
- Massive data management and analysis.
- Perpetual connectivity.

IoT Applications



11. Tactile Internet



Tactile Internet

- Tactile Internet is considered as the next generation of IoT networks.
- It will open up new domains of tactile Internet services such as remote surgery, robotics, autonomous vehicles and many more.
- The first telesurgical operation was carried out at 2001. However, telesurgery is still not in the mainstream due to the technical limitations.
- In industrial environments, it can be used for remote mining in high-risk areas.

Tactile Internet : Surgery



Thank You